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TURBULENT SWIRLING FLAMES: EXPERIMENTAL INVESTIGATION OF THE FLOW FIELD AND FORMATION OF NITROGEN OXIDE

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The NO emissions of a 150 kW natural gas (with and without fuel N) burner were shown to be dependent on the geometry of the gas nozzle and the swirl intensity of the combustion air. Numerous detailed in-flame measurements were performed in order to understand the local conditions that influence the formation of NO in highly turbulent swirling flames. It was found, that by controlling the axial fuel gas momentum in combination with the swirl number, mixing of fuel and air can be influenced, thus reducing NO formation. Using fuels with chemically fixed nitrogen requires flames with the main heat release under fuel-rich conditions, leading to type I flames. These flames have the potential to reduce the NO emissions up to 70% compared with premixed flames. To minimize thermal NO formation, high-temperature zones with near-stoichiometric mixtures must be avoided, which may be realized by type II flames. Moreover, swirl turned out to be a strong tool for minimizing thermal NO. Under overall fuel-lean conditions, higher swirl led to faster mixing of fuel and air, thus lowering the temperature level of the flame and reducing thermal NO emissions. The experiments were performed in cooperation with the German Technische Flammen (TECFLAM) joint research program, using a standardized combustion chamber and burner. By applying identical experimental setups and different measurement techniques by the TECFLAM partners, accurate and reliable measuring data were produced on the time-mean and on the turbulent fluctuation levels.

Introduction

Swirl-stabilized non-premixed flames are common in technical combustion devices such as industrial furnaces, incinerators, and gas turbines. These flames offer the possibility to control the flow and mixing pattern and thus improve ignition stability and minimize NO_x emissions by adapting the aerodynamic inlet conditions. Furthermore, the flame shape can be adjusted to the geometry of the combustion chamber.

A comprehensive understanding of the NO formation in these rather complex turbulent flames can be attained by taking systematic in-flame measurements of chemical species, flow velocities, and temperatures. In-flame measurements of enclosed swirling flames which were reported in the literature [1–3] typically comprise measurements of either the flow field, the temperature, or the species concentrations. However, for the interpretation of NO formation, all local information, including residence time, stoichiometry, and temperature, must be analyzed. Furthermore, it is important to include or focus on the burner near field as the main reaction zone, with high temperatures located there. Complete sets of data from semi-industrial swirling flames including inlet and boundary conditions are also useful and necessary for validation and improvement of numerical models.

This work provides detailed data from the flow field, temperature, and chemical species of a 0.15 MW non-premixed, highly turbulent swirling flame. The experiments were performed in a joint research project of the Technische Flammen (TECFLAM) Group, which is a consortium with research groups from the University of Karlsruhe, University of Darmstadt, University of Heidelberg, University of Stuttgart, and DLR Stuttgart.

Experimental Setup

The experiments were carried out using the TECFLAM combustion chamber (see Fig. 1). This cylindrical, water-cooled combustion chamber ($T_{\mathrm{Wall}} \approx$ 150 °C) has an inner diameter of 0.5 m and a length of 1.7 m. To perform field measurements, the burner can be moved axially inside the combustion chamber. The boundary and inlet conditions during the experiments were controlled and kept constant to guarantee reproducibility of the measurements.

The swirling flow was induced by a movable-block swirl generator [4] integrated into the burner, that allowed a continuous adjustment of the theoretical swirl number between $S_{0,\mathrm{th}} = 0.0$ (no swirl) and 2.0 (full swirl). The diameter of the burner outlet, D, was 60 mm. Different kinds of quarls can be

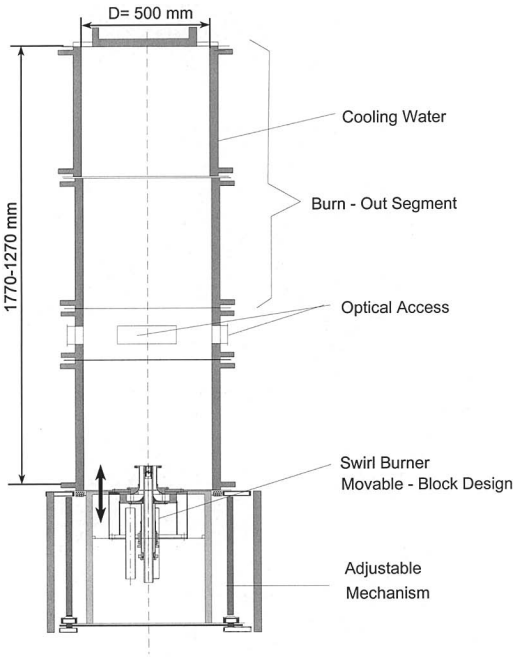

FIG. 1. TECFLAM combustion chamber with standard swirl burner.

mounted onto the burner mouth. In diffusion mode, the fuel gas was injected concentrically through exchangeable gas nozzles (multihole, annular channel, and single-hole nozzles). The burner can also be operated in premixed mode. For this purpose, the fuel gas was injected either immediately downstream of the swirl generator or preferably upstream of the swirler through eight nozzles to provide complete mixing in the burner exit plane. To achieve similar inlet conditions compared with the diffusion mode, a cylindrical bluff body was installed instead of the nozzle.

The measurements in the near field of the burner were the turbulent velocity and temperature fields and the fields of concentrations of relevant stable gas species. Velocity measurements were carried out using a two-channel laser Doppler velocimeter. The measurements included the determination of the mean velocities, U, V, and W, their fluctuation values, u', v', and w', and the Reynolds stresses, $u'v'$ and $u'w'$. Magnesia oxide particles ($d_P = 1$–$5\ \mu$m) brushed off from a massive cylinder served as tracer material for the combustion gas and air. Time-mean and fluctuating temperatures were measured using thermocouples with diameters of 50 μm. After preamplification, the thermal inertia of the thermocouples was compensated for electronically, so that temperature fluctuations up to about 1 kHz could be detected. Concentrations of stable species were determined by using a water-cooled probe and analyzing the gas samples with commercial gas analyzers operating on the basis of infrared absorption (CH_4, CO, CO_2, H_2O), paramagnetism (O_2), and thermal conductivity (H_2). Fourier transform infrared was used to determine the concentrations of HCN, NH_3, C_2H_2, C_2H_4, and H_2O. The analyzer tube was heated electrically to minimize heat losses and to avoid condensation effects. The probe operated isokinetically, with the probe tip oriented toward the main stream direction. The comparison of the measurements with results from other research groups (velocities, University of Darmstadt; concentrations of chemical species, DLR Stuttgart and University of Heidelberg [5]) showed excellent agreement.

Results and Discussion

Flow and Mixing Pattern

The highly turbulent flames were stabilized by inner recirculation zones. These zones can be induced by a bluff body, swirl, or both. The stability of these flames is favored over the entire range of operation by the following conditions in the ignition zone: near-stoichiometric mixture, low mean flow velocities, energy transport of heat and chemically active species by turbulent exchange from the reverse flow to ignite the fresh mixture, and high turbulence intensities. Swirl and disk stabilized flames can be subdivided, by different mixing and reaction pattern, into two types of flames [4] (Fig. 2):

1. The central fuel jet penetrates the inner recirculation zone completely (compare with the flow field in Fig. 10). Then the internal reverse flow builds a small annular recirculation zone between the fuel jet and combustion air. This type of flame is denoted as a type I flame.
2. When the internal reverse flow strength prevents the penetration of the recirculation zone by the fuel jet, the fuel jet is bent toward the annular shear layer. Because of the rapid mixing in the shear layer, this flame type is very stable and short. This type of flame is denoted as a type II flame (compare with the flow field in Fig. 7).

To illustrate the different flow patterns, measurements of the axial velocity along the axis for different gas nozzle outlet diameters, d, are shown in Fig. 3. For gas outlet velocities higher than 34 m/s ($d <$ 12.5 mm), the jet is able to penetrate the inner recirculating zone, which has a length of $x/D \approx 1.2$ (type I flame). Jets with a lower outlet velocity are not able to penetrate the inner reverse flow completely (type II flame). The formation of these flame types also depends on the swirl number, burner mouth geometry (position of the gas gun), and air outlet momentum [6].

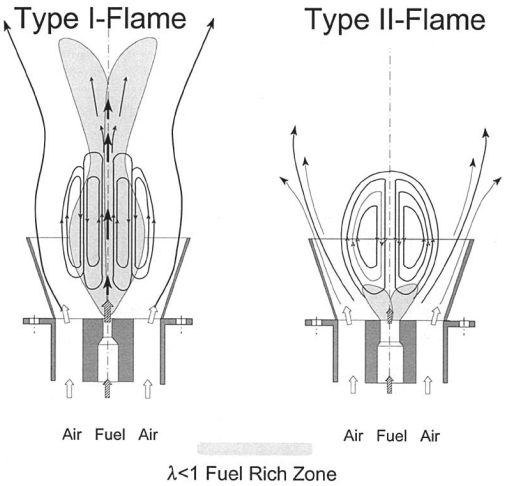

FIG. 2. Flow and mixing fields of type I and type II swirling flames.

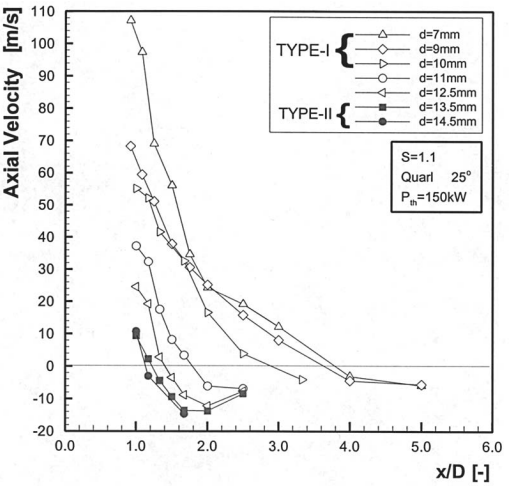

FIG. 3. Axial velocities along the centerline for different gas outlet diameters, d.

The inner recirculating mass flow rates of type I and type II flames as a function of the axial burner distance are given in Fig. 4; those data were obtained by cross section integration of measured axial velocity and temperature (density) profiles. The recirculation of heat and radicals with the central reverse flow ignited the unburned gas in the reaction zone. Basically, higher swirl intensity led to lower pressure at the axis and therefore to higher recirculated mass flow rates. The combination of axial gas injection and an additional diverging burner quarl led to high recirculated mass flow rates, with the consequence of excellent ignition stability also of type I flames.

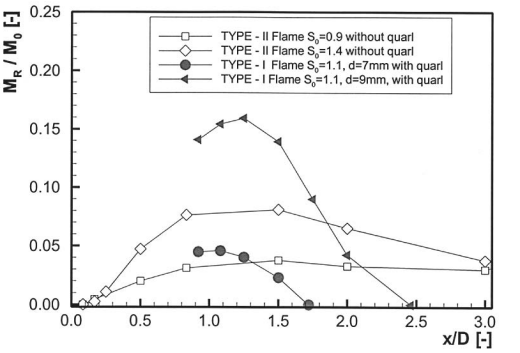

FIG. 4. Recirculated mass flow rates of type I and type II swirling flames.

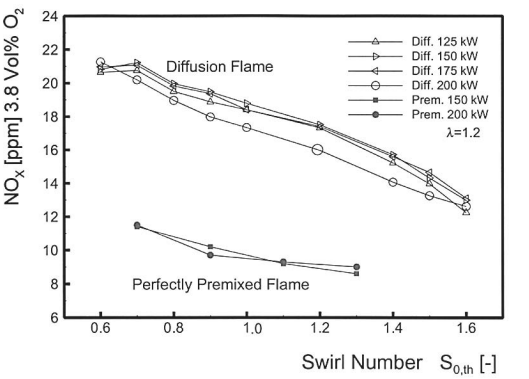

FIG. 5. NO_x emissions of type II flames with different swirl numbers ($S_{0,th}$).

Fields of Concentrations of Chemical Species and Temperatures

In the following, the formation of nitrous oxides in high-intensity type II swirling flames with natural gas as fuel is discussed. The main objectives of this study were to answer the question where and under which conditions (local temperature and gas composition) nitrous oxides are being formed in swirling flames. In Fig. 5, the total NO concentrations based on 3.8 vol % O_2 for type II flames at a total air equivalence ratio of $\lambda = 1.2$ for different thermal throughputs and swirl numbers are illustrated. In diffusion mode, the fuel gas injection was realized by an annular ring nozzle. To maintain similar outlet geometries, in premixed mode the gas nozzle was replaced by a central blockage. The results demonstrate that the emission of NO decreases with increasing swirl number, independently of the thermal throughput. The same tendencies exist for premixed flames, where due to lower temperatures in the main reaction zone the overall values of NO concentrations are lower. The radial profiles of the mean temperatures at two different axial burner distances (x/

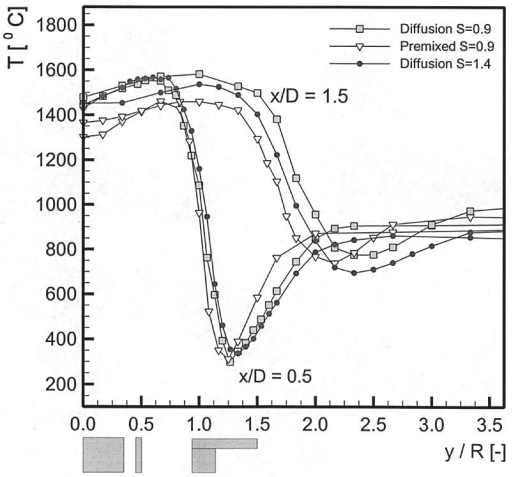

FIG. 6. Radial profiles of the mean temperature for two axial burner distances.

$D = 0.5$ and 1.5) displayed in Fig. 6 confirm these findings. For diffusion flames, the highest temperatures were obtained in the main reaction zone, located at the boundary of the inner recirculation zone. With increasing swirl ($S = 0.9 \rightarrow 1.4$), the entrainment of the jet also increased (see Fig. 4), and consequently temperatures further downstream decreased. In the inner backflow zone close to the axis, where mainly burnout gases prevail, a low heat loss (no convective heat loss to the chamber wall) led to high temperatures. The maximal temperatures between premixed and diffusion flames differed by 100 to 180 K, thus explaining higher thermal NO emissions of diffusion flames.

In Fig. 7, the fields of axial velocities, temperatures, and mixture fraction as well as local NO_x concentrations of two diffusion flames at different swirl numbers ($S = 0.7$ and 1.4) are compared. The experiments were performed at a total air equivalence ratio of 1.2 and a thermal throughput of 150 kW. In the bottom of the diagram, the burner outlet with an annular ring gas nozzle is indicated for clarity. The comparison of the average axial velocities, U, shows, especially along the axis, that higher swirl led to a higher backflow rate. However, an increasing backflow velocity led to lower residence times in the recirculation zone and a higher recirculation of burned gas. The level of total NO_x emission of flame configurations with high swirl numbers was much lower than that with lower swirl numbers ($S = 0.7$). Maximum NO concentrations appeared in the inner recirculation zone. Flames with high swirl numbers and therefore lower residence times due to higher backflow velocities tended to form less thermal NO in the inner recirculation zone. In addition, higher swirl caused faster mixing and therefore lower mean temperatures in the main reaction zone, with lower NO emissions. Measurements from the same TECFLAM standard flame performed at PCI Heidelberg applying the measurement technique of laser-induced fluorescence delivered comparable results [5].

Another means to affect NO emission, especially of type I flames, is the variation of the axial momentum of the fuel jet. In Fig. 8, the NO_x emissions for type I flames are given as a function of the air equivalence ratio. In general, all curves can be divided into two sections: overall air equivalence ratios higher than 1 (fuel-lean conditions) and less than 1 (fuel-rich and near-stoichiometric mixture). At an air equivalence ratio close to 1, the perfectly premixed flame exhibited the highest concentration of nitrogen oxides. The emission dropped significantly toward lower total air equivalence ratios. In this section, the emissions were almost independent of the geometry of the fuel gas supply. In contrast, under fuel-lean conditions (air equivalence ratio higher than 1), a strong dependency of the NO emissions on the gas nozzle geometry could be observed. With increasing gas nozzle diameter and thereby decreasing axial momentum of the fuel jet, the emissions of nitrous oxides decreased. Confirmed detailed in-flame measurements show that small fuel outlet diameters led to an intense burnout of fuel in the jet zone of the flame at long residence times at a high temperature level. By lowering the momentum of the fuel jet, the emissions dropped significantly. Type II swirl flames are also illustrated in Fig. 8. At high air equivalence ratios, the lowest values of nitrous oxides were being emitted from the perfectly premixed flame.

Nitrogen Fuel Conversion

The emission of nitrogen oxides from technical combustion devices often is dominated by the conversion of nitrogen contained in the fuel. Chemically fixed nitrogen is converted during combustion largely to ammonia, which then is transformed to nitrogen oxides. To investigate the conversion of chemically fixed nitrogen, therefore, the fuel was doped with ammonia.

The formation of nitrogen oxides from chemically fixed nitrogen is highly dependent on the local stoichiometry. As in type II flames, the main reaction takes place under fuel-lean conditions, the reaction of fixed nitrogen to NO instead of N_2 is preferred, and high emissions of total fixed nitrogen ($[TFN] = [NO] + [NO_2] + [NH_3] + [HCN]$) result. Flames with internal aerodynamically staged mixing of fuel and air and therefore large fuel-rich zones offer the possibility to decrease NO emissions considerably.

To minimize NO_x emissions from turbulent swirling flames with NH_3 dotted natural gas as fuel, detailed in-flame measurements and parametric studies for both flame types were performed. Results of

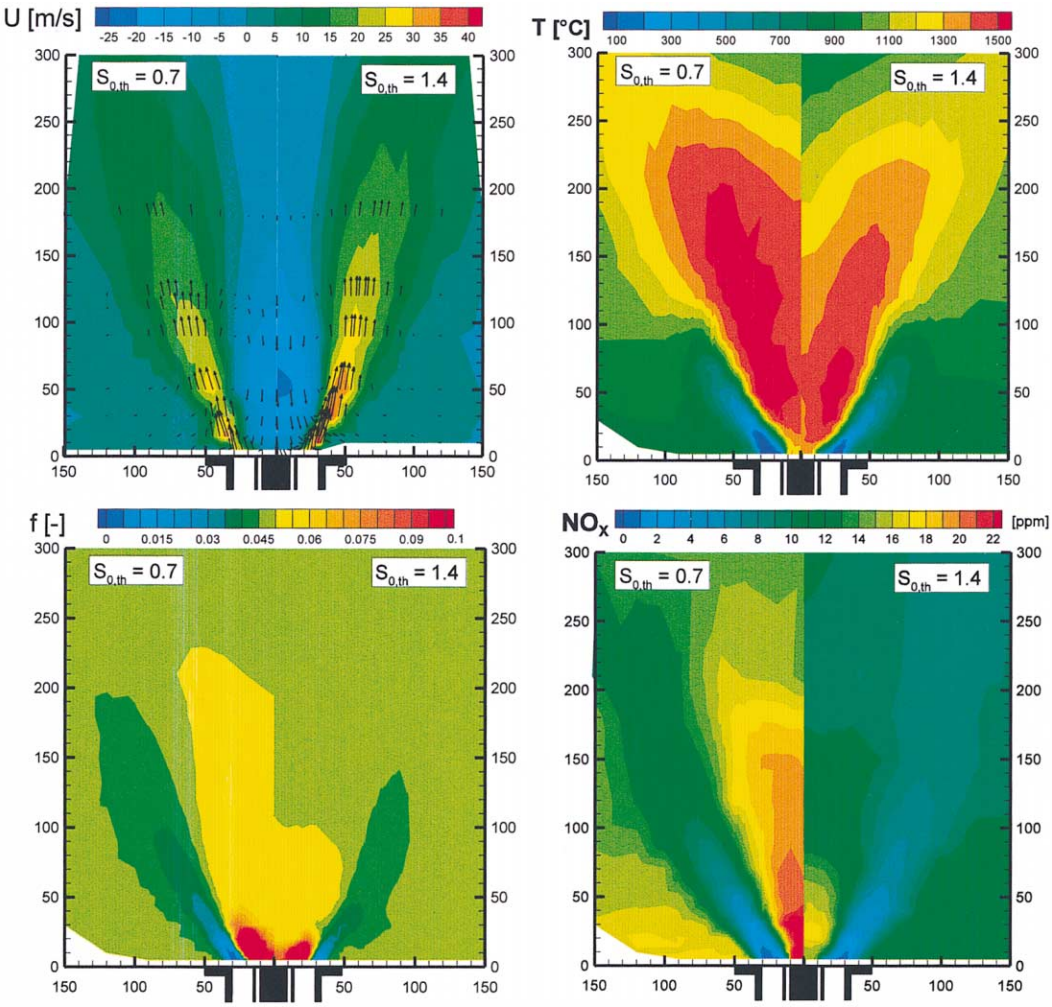

FIG. 7. Fields of axial flow velocity (U), temperature (T), mixture fraction (f), and NO_x concentrations of two type II flames with different swirl numbers ($S_{0,\mathrm{th}} = 0.7$ and 1.4); $P_{\mathrm{th}} = 150$ kW; $\lambda = 1.2$.

the exhaust gas analysis are given in Fig. 9. It is obvious that the levels of TFN emission from premixed and diffusion type II flames were highest. NO emissions from type I flames were much lower over the complete range of the air equivalence ratio tested. The typical curve shape of [TFN] versus λ of type I flames could be split into a fuel-rich and a fuel-lean part [7]. Because of a lack of oxygen under fuel-rich conditions, the oxidation of NH_3 and HCN was not complete, and these compounds were measured in the exhaust. On the other hand, under fuel-lean conditions, the high concentrations of oxygen favored the formation of NO.

The fuel jet momentum was varied to try to minimize NO_x emissions; it became obvious that the mechanisms behind this dependence are quite complex, so that field measurements were indispensable. As an example, the results of the field measurements of a type I flame with a nozzle diameter of 7 mm are discussed. The operating conditions for that run were a thermal throughput of 150 kW, a total air equivalence ratio of 1.2, a swirl number of 1.1, and as fuel natural gas dotted with 3 vol % NH_3. In Fig. 11, the mixture fraction calculated with the help of the concentrations of the stable chemical species illustrates fuel-rich conditions over a wide range, especially in the inner recirculation zone and in the jet close to the axis. The ignition zone is located in the reverse flow, indicated by high temperatures (see Fig. 10). In the fuel-lean region near the air flow, NO formation proceeds rapidly; see Fig. 11 (right side), where calculated NO formation rates are plotted. These calculations were performed with the measured temperatures and concentrations of stable

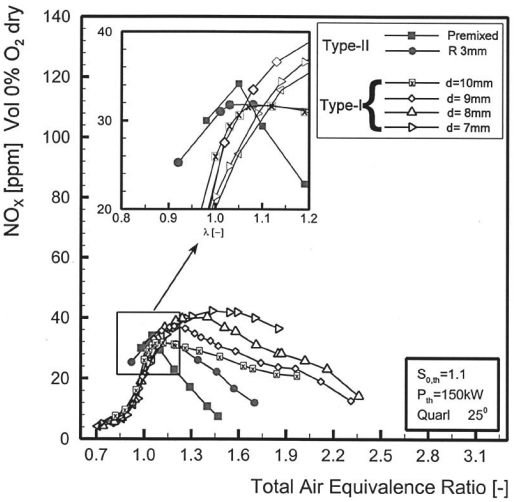

FIG. 8. NO_x emissions of type I flames with different axial fuel gas momentum (d = fuel gas outlet diameter).

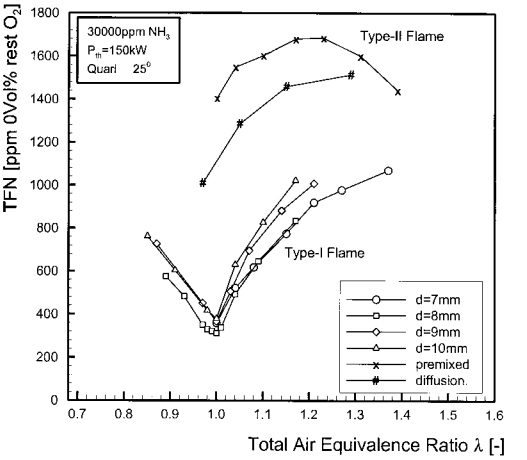

FIG. 9. TFN emissions from swirling flames with 3 vol % NH_3 mixed in natural gas.

species as input conditions, using the software package SENKIN in combination with the modified Gas Research Institute (GRI) mechanism [8]. On the fuel-rich side in the inner recirculation zone, however, NO consumption by NO reburning can be observed. The comparison between the mixture fraction and the calculated NO formation rates shows that the production and consumption of NO correlate directly with the local mixture [9]. Because of insufficient residence times in the jet zone, reburning effects there are negligible, and therefore the main production of NO takes place in the jet of the flame. As a result, minimization of NO_x emissions in these flames is obtained by a maximization of recirculated mass flow in the near burner field under fuel-rich conditions and minimization of heat release in the jet zone. This may be performed by a modified burner geometry—for example, length and shape of the quarl in combination with optimized flame input conditions of swirl intensity and axial momentum.

Conclusions

Detailed in-flame measurements of flow field, temperatures, and gas composition for type I and type II swirling flames were performed to clarify the

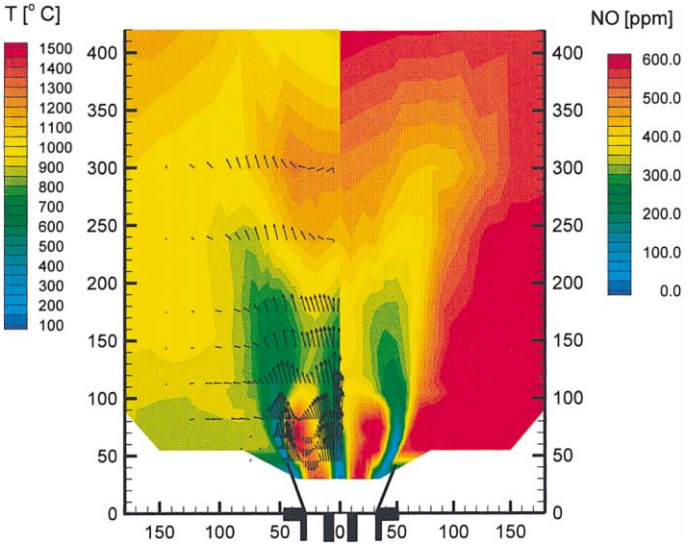

FIG. 10. Field of temperature (T) and NO concentrations of a type I flame (3 vol % NH_3 in natural gas; λ = 1.2; P_{th} = 150 kW).

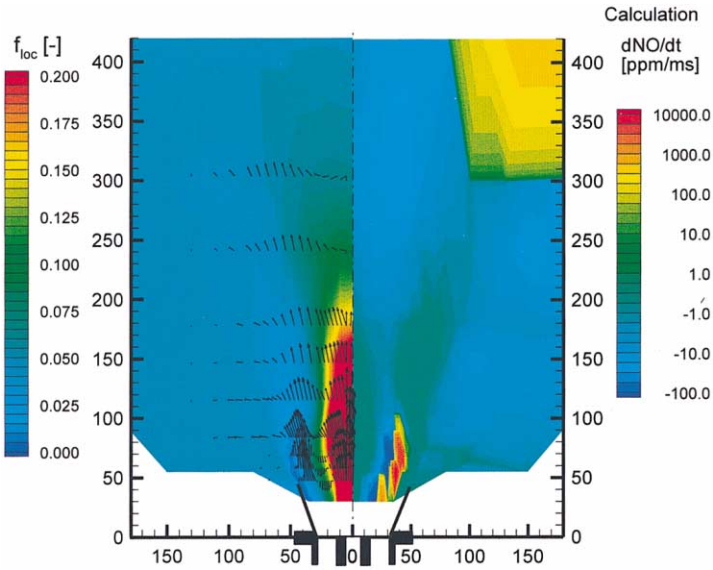

FIG. 11. Field of mixture fraction (f) and calculated NO formation rates of a type flame (3 vol % NH_3 in natural gas; $\lambda = 1.2$; $P_{th} = 150$ kW).

mechanism of NO formation for a further minimization of NO emissions. The following conclusions can be drawn:

1. For type-II swirling natural gas flames, an increasing swirl intensity leads to a higher recirculated mass flow rate through the inner recirculation zone, with high temperatures. As the residence time in that zone decreases, with increasing recirculated mass flow, NO formation is also reduced.
2. By varying the axial momentum of the central fuel jet, it is possible to control the ratio of heat release between jet and inner recirculation zone. An increasing heat release in the jet of the flame causes an increase of total NO_x emissions, because of long residence times in high-temperature zones. On the other hand, the fuel-rich conditions in the heat release zone of these flames significantly reduce NO_x emissions from fuels with chemically fixed nitrogen.

Acknowledgments

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